AWS S3 Lifecycle Management and Replication

Practical Cloud Experiment Report

Primary Region: ap-south-1 (Mumbai)
Additional Regions: Singapore and Ohio
Experiment: Lifecycle rules and S3 replication

1 Objective

The objective of this experiment is to understand how Amazon S3 manages object lifecycles and how objects can be replicated between buckets.

The experiment covers the following tasks:

- Configure lifecycle rules to manage old object versions, expired delete markers, and incomplete multipart uploads.
- Configure a transition rule for eligible objects to move to S3 Standard-Infrequent Access (Standard-IA).
- Configure the IAM permissions and role required for S3 replication.
- Explore same-account replication within the same Region and across different Regions.
- Explore cross-account replication within the same Region and across different Regions.
- Verify replication by checking the destination buckets and replication status where available.

2 Description

Amazon S3 is an object storage service. Objects are stored in buckets, and each bucket is associated with an AWS Region.

Two S3 capabilities were studied in this experiment.

2.1 Lifecycle Management

Lifecycle rules automate actions on objects over time. Depending on the rule configuration, S3 can expire noncurrent object versions, remove expired object delete markers, abort incomplete multipart uploads, or transition eligible objects to another storage class.

Lifecycle rules do not necessarily execute immediately after being enabled. Eligibility, object size, and S3's processing schedule affect when an action occurs.

2.2 Replication

S3 replication automatically copies eligible objects from a source bucket to a configured destination bucket.

- **SRR:** Same-Region Replication.
- **CRR:** Cross-Region Replication.
- **Same-account replication:** Source and destination buckets belong to the same AWS account.

- **Cross-account replication:** The destination bucket belongs to another AWS account and requires suitable destination permissions.

Replication is asynchronous. A successful configuration does not mean that every object will appear in the destination immediately.

For this experiment, the source object used for testing was `text.txt`, approximately 35 bytes in size, with SSE-S3 encryption.

3 Architecture and Resources

3.1 Architecture Diagram

The architecture is divided into two main components: **S3 Lifecycle Management** and **S3 Replication**. The lifecycle component operates on the source bucket by applying predefined rules to manage object versions and storage usage. The 'Test-Rule' handles noncurrent-version expiration, expired delete-marker removal, and incomplete multipart upload cleanup. The 'test-rule-transition' rule is configured to move eligible objects to S3 Standard-IA after one day, subject to the configured minimum object-size requirement of 128 KB. These rules demonstrate how storage management tasks can be automated instead of being performed manually for every object.

The replication component demonstrates how an object can be copied from a source bucket to different destinations based on the required storage architecture. The source bucket is located in the Mumbai Region ('ap-south-1') under AWS account '715615725174'. From this source, the experiment covers four scenarios: same-account, same-Region replication; same-account, cross-Region replication to Singapore; cross-account, same-Region replication; and cross-account, cross-Region replication to Ohio. This arrangement provides a practical comparison of how the destination account and Region affect the replication configuration.

IAM permissions connect the replication workflow. The IAM user configures the replication rules and passes the 'S3-Replicator' role to Amazon S3. The role's trust policy allows the S3 service to assume it, while its permissions policy defines the source-reading and destination-replication operations it can perform. For cross-account replication, the destination bucket policy must also authorize the source account's replication role. Both source and destination buckets require versioning to be enabled, and replication should be verified by checking the replication status and the destination object. Since replication and lifecycle processing are asynchronous, the architecture represents the configured workflow rather than a guarantee that every action completes immediately.

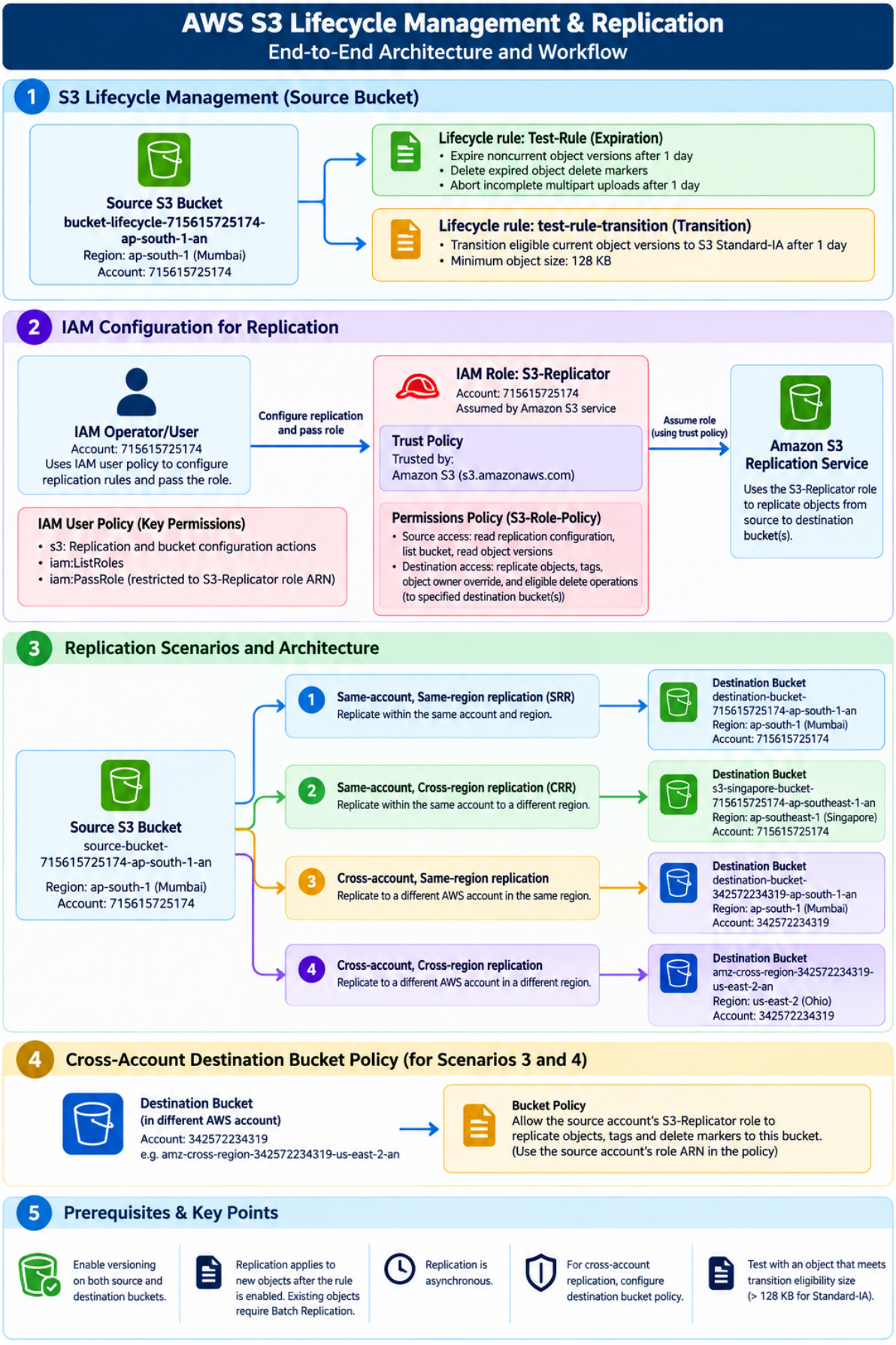


Figure 1: Overview of S3 lifecycle management and replication

3.2 Bucket Inventory

Bucket	Account	Region	Purpose
bucket-lifecycle-715615725174 -ap-south-1-an	715615725174	ap-south-1	Lifecycle rules
source-bucket-715615725174 -ap-south-1-an	715615725174	ap-south-1	Replication source
destination-bucket-715615725174 -ap-south-1-an	715615725174	ap-south-1	Same-account SRR destination
s3-singapore-bucket-715615725174 -ap-southeast-1-an	715615725174	ap-southeast-1	Same-account CRR destination
destination-bucket-342572234319 -ap-south-1-an	342572234319	ap-south-1	Cross-account SRR destination
amz-cross-region-342572234319 -us-east-2-an	342572234319	us-east-2	Cross-account CRR destination

The second account, 342572234319, is the destination account used for the cross-account scenarios.

4 Part A: S3 Lifecycle Management

4.1 Lifecycle Bucket

The lifecycle experiment was performed on the bucket

bucket-lifecycle-715615725174-ap-south-1-an

The bucket is in the Mumbai Region, ap-south-1, under AWS account 715615725174.

Two lifecycle rules were configured.

4.2 Rule 1: Test-Rule

The rule named Test-Rule contains the following actions:

1. Permanently expire noncurrent object versions after one day.
2. Delete expired object delete markers.
3. Abort incomplete multipart uploads after one day.

These actions help manage old object versions and unfinished multipart uploads. The noncurrent-version expiration setting concerns previous versions of objects in a versioned bucket; it does not mean that every current object is deleted after one day.

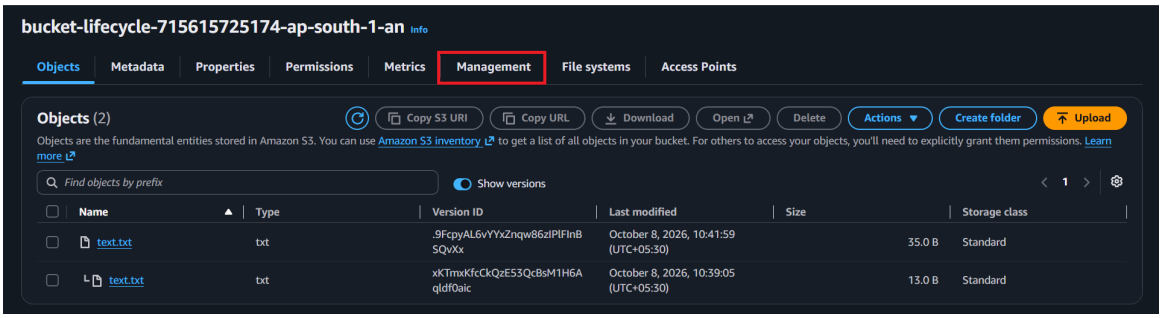


Figure 2: Lifecycle bucket and versioning overview

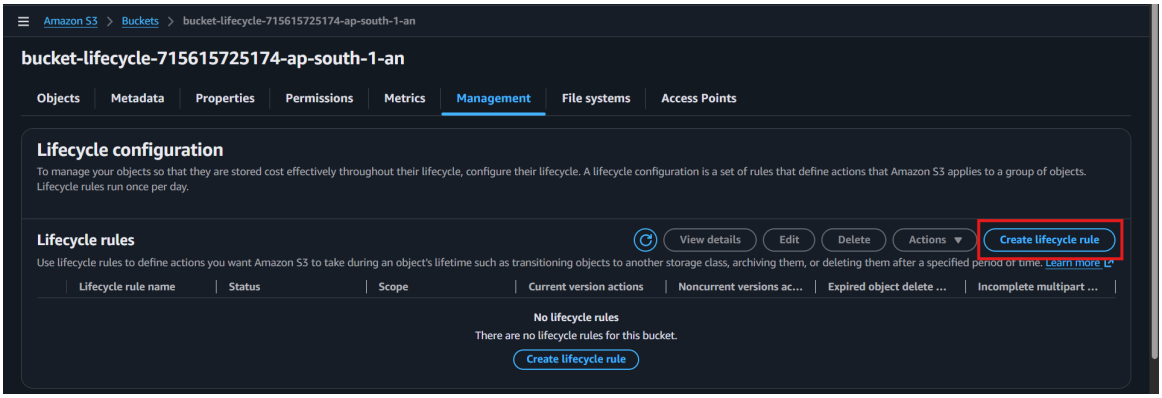


Figure 3: Starting the lifecycle rule configuration

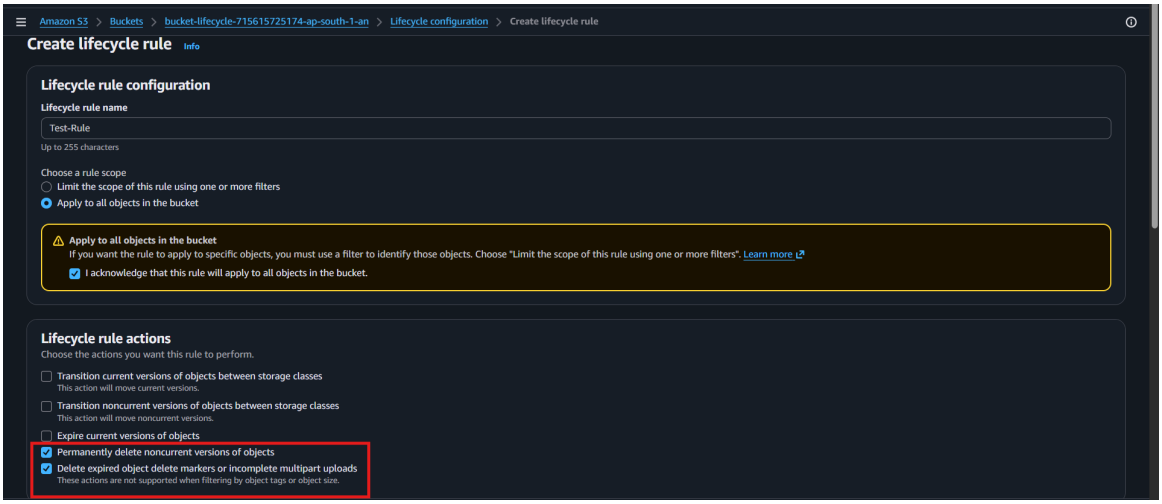


Figure 4: Configuration of the Test-Rule lifecycle rule

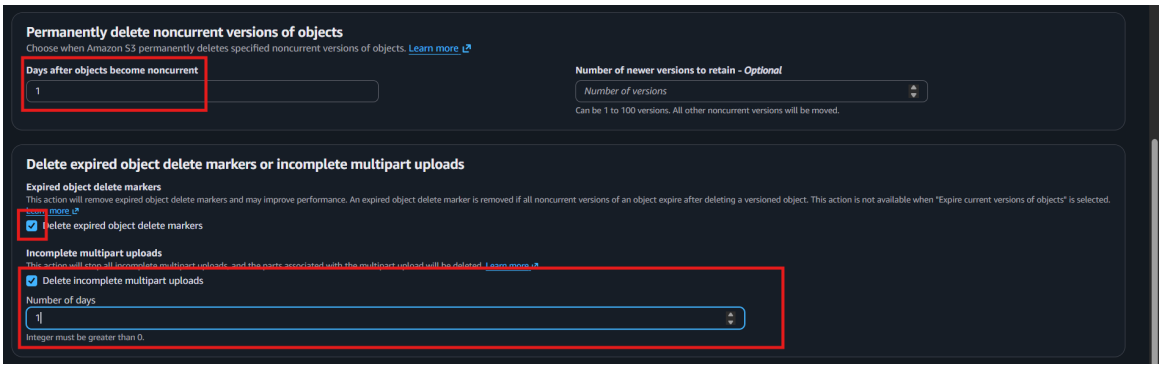


Figure 5: Noncurrent-version expiration and incomplete multipart upload settings

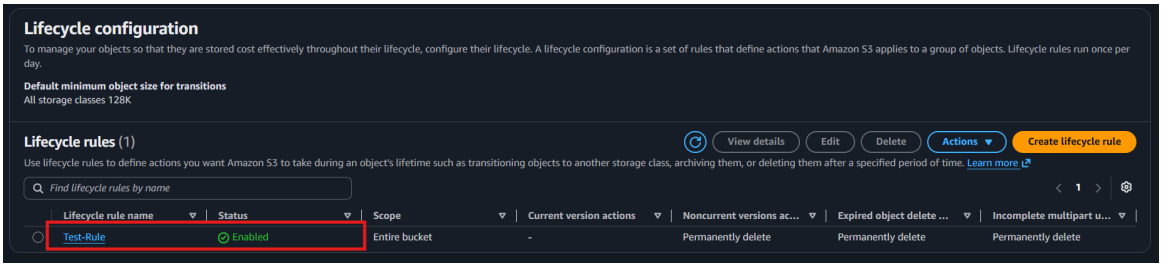


Figure 6: Test-Rule displayed as enabled

4.3 Rule 2: test-rule-transition

The second rule is named `test-rule-transition`. It is configured to transition eligible current object versions to S3 Standard-IA after one day.

The configured minimum object size is 128 KB.

The test object `text.txt` was approximately 35 bytes. Therefore, it is smaller than the configured minimum object size. The configuration alone does not establish that this object transitioned to Standard-IA.

This distinction is important: a lifecycle rule can be enabled while a particular object remains ineligible for the configured transition.

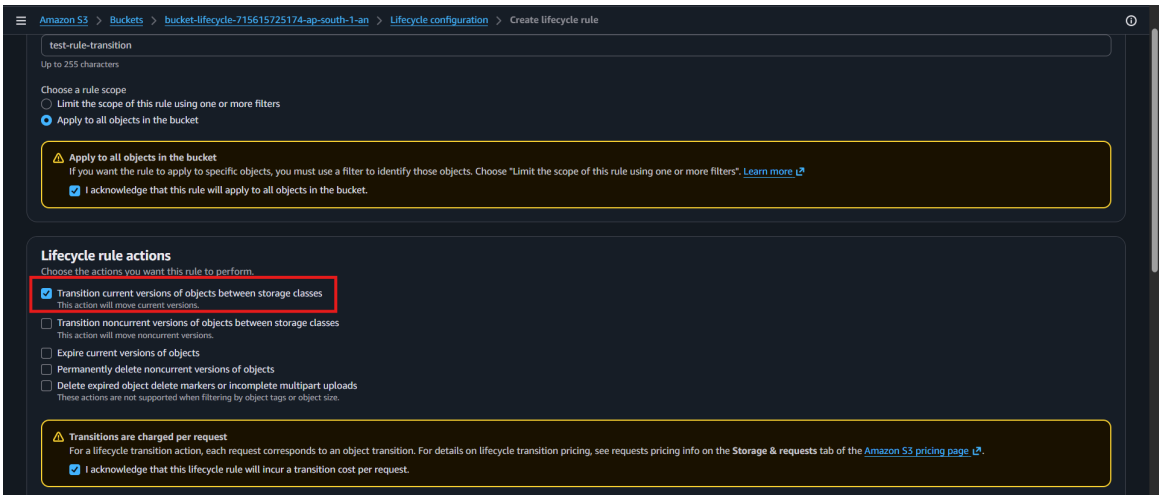


Figure 7: Creating the transition lifecycle rule

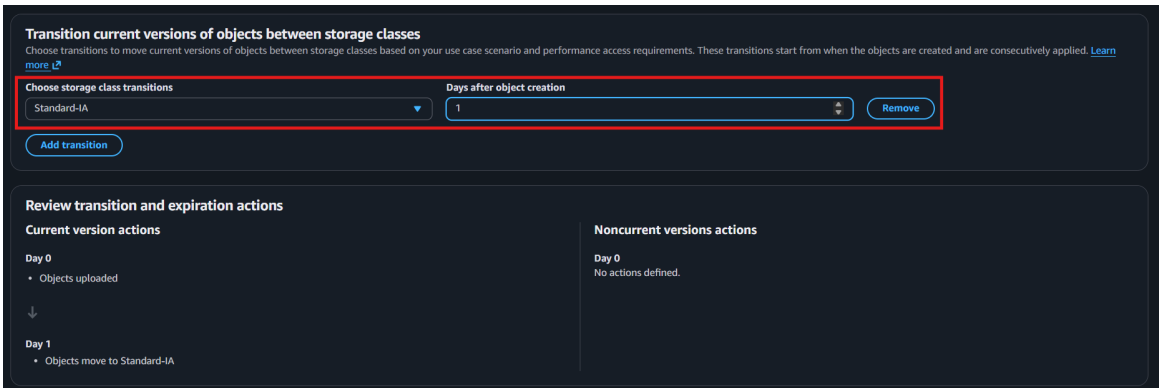


Figure 8: Transition to Standard-IA after one day and the minimum object-size setting

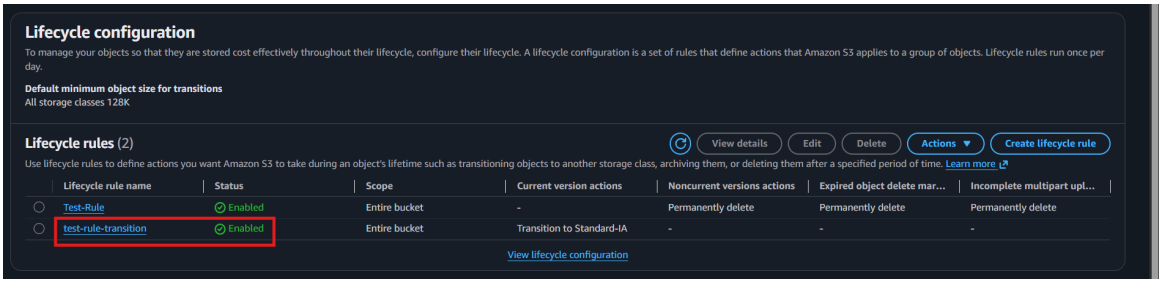


Figure 9: Lifecycle rules displayed in the bucket

4.4 Lifecycle Verification Notes

The lifecycle configuration should be checked against the rule settings in the S3 console.

- Confirm that both rules are enabled.
- Confirm the one-day expiration and multipart-upload settings in `Test-Rule`.
- Confirm the Standard-IA transition and minimum-size setting in `test-rule-transition`.
- Do not infer that an action completed merely because the corresponding rule is enabled.
- Check an object's storage class and version information separately when verifying an actual lifecycle action.

5 Part B: IAM Configuration for Replication

S3 replication requires permissions for the user configuring the replication rule, permissions for the role assumed by S3, and, for cross-account replication, suitable permissions on the destination bucket.

5.1 IAM User Permissions

The IAM user policy contains permissions used to configure S3 resources, lifecycle rules, replication configuration, and to pass the replication role to the service.

The relevant permissions present in the supplied policy include:

- `s3:GetLifecycleConfiguration`
- `s3:PutLifecycleConfiguration`
- `s3:GetReplicationConfiguration`
- `s3:PutReplicationConfiguration`
- `iam:ListRoles`
- `iam:PassRole`

The `iam:PassRole` permission is restricted to the specific `S3-Replicator` role ARN.

5.2 S3-Role-Policy Permissions

The customer-managed permissions policy attached to the `S3-Replicator` role is named `S3-Role-Policy`. It contains permissions for accessing the source bucket and replicating objects to the configured destination.

The main permissions are:

- **Source access:** `s3:GetObjectVersionAcl`, `s3:ListBucket`, `s3:GetObjectVersionForReplicati` `s3:GetBucketVersioning`, `s3:GetReplicationConfiguration`, and `s3:GetObjectVersion`.
- **Destination access:** `s3:ReplicateObject`, `s3:ReplicateDelete`, `s3:ReplicateTags`, and `s3:ObjectOwnerOverrideToBucketOwner`.

The source permissions allow the role to access the required bucket configuration and object-version information. The destination permissions support object replication, tag replication, eligible delete-marker replication, and destination-owner override when configured.

The supplied policy specifies the source bucket `source-bucket-715615725174-ap-south-1-an` and the destination object ARN for `amz-cross-region-342572234319-us-east-2-an` in Ohio. Therefore, the destination permissions in this policy should not automatically be assumed to cover the other replication destinations.

The complete policy JSON, including its resource ARNs, is provided in Appendix B.

5.3 Replication Role

The IAM role is named `S3-Replicator`.

Its trust relationship allows the Amazon S3 service to assume the role. Its permissions policy is named `S3-Role-Policy`.

These are separate concepts:

- **Trust policy:** Determines which principal can assume the role. For this experiment, the trusted service is Amazon S3.

- **Permissions policy:** Defines the source-reading and destination-replication actions permitted to the role.

The role policy includes permissions to read the source bucket's replication-related information and object versions, and to replicate objects, tags, delete markers, and ownership overrides to the specified destination objects.

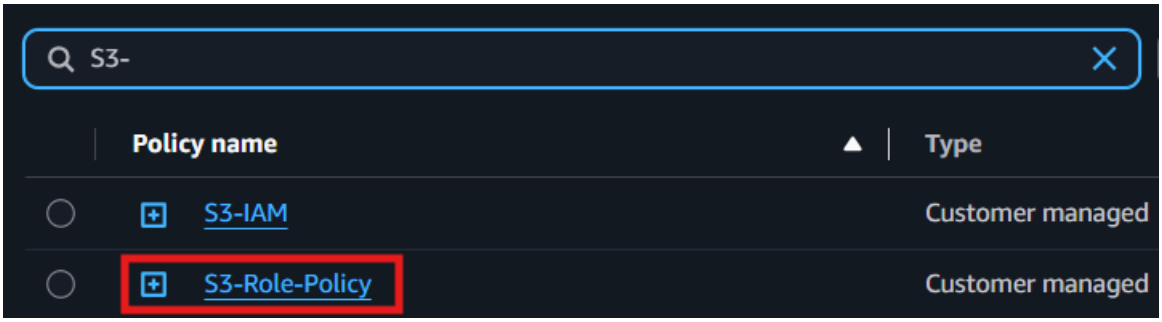


Figure 10: Existing customer-managed S3 role policy

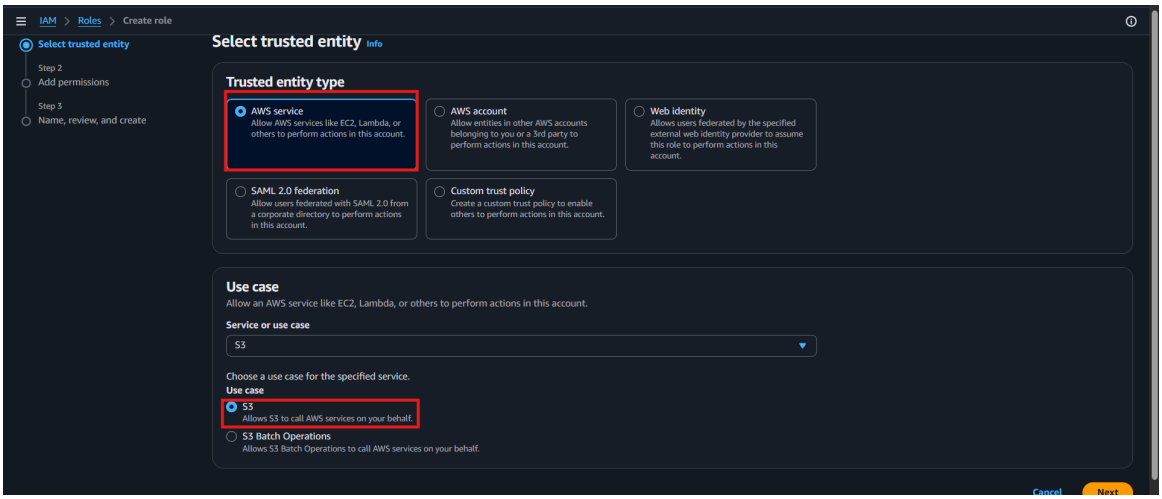


Figure 11: Selecting Amazon S3 as the trusted service

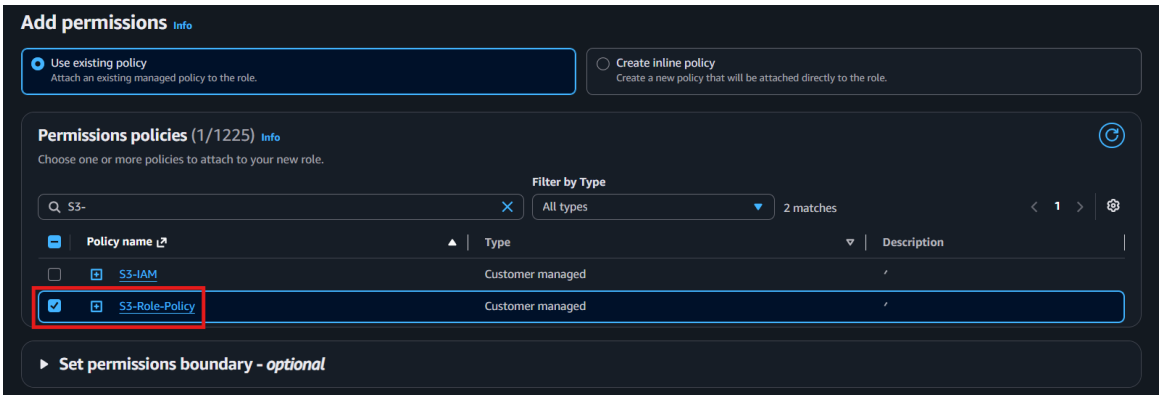


Figure 12: Attaching the permissions policy to the role

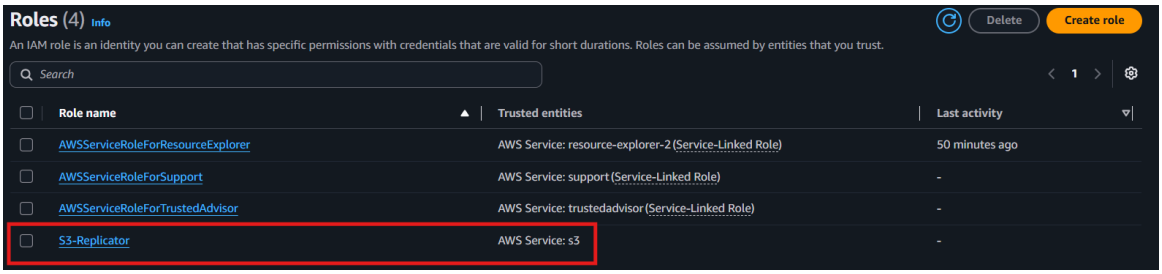


Figure 13: S3-Replicator role

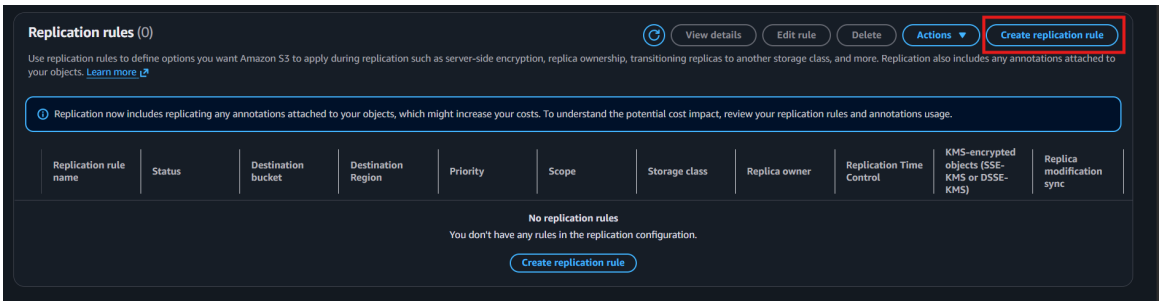


Figure 14: Replication configuration before completing the rule setup

5.4 Important Resource Scope

The supplied role policy references the Ohio cross-account destination bucket in its destination resource ARN. That does not automatically authorize replication to the other destinations.

For another replication rule, the destination resource permissions must match the intended destination bucket. In general:

- Bucket-level actions such as `s3:ListBucket` use the bucket ARN.
- Object-level actions such as `s3:ReplicateObject` use the object ARN, commonly ending in `/*`.
- `iam:PassRole` uses the role ARN.

The policy JSON in the appendix records the supplied configuration. It should not be treated as a universal policy for every destination.

6 Part C: Replication Scenarios

The replication source bucket used for the scenarios is

`source-bucket-715615725174-ap-south-1-an`

It belongs to account `715615725174` and is located in `ap-south-1` (Mumbai). Versioning must be enabled on both the source and destination buckets.

The experiment covered four replication patterns.

6.1 Scenario 1: Same-Account SRR

The source and destination buckets are in the same account and Region.

- Source: `source-bucket-715615725174-ap-south-1-an`
- Destination: `destination-bucket-715615725174-ap-south-1-an`
- Account: `715615725174`
- Region: `ap-south-1` (Mumbai)

The destination bucket is in the same account and the same Region as the source. The replication rule is configured to copy eligible new objects to the destination.

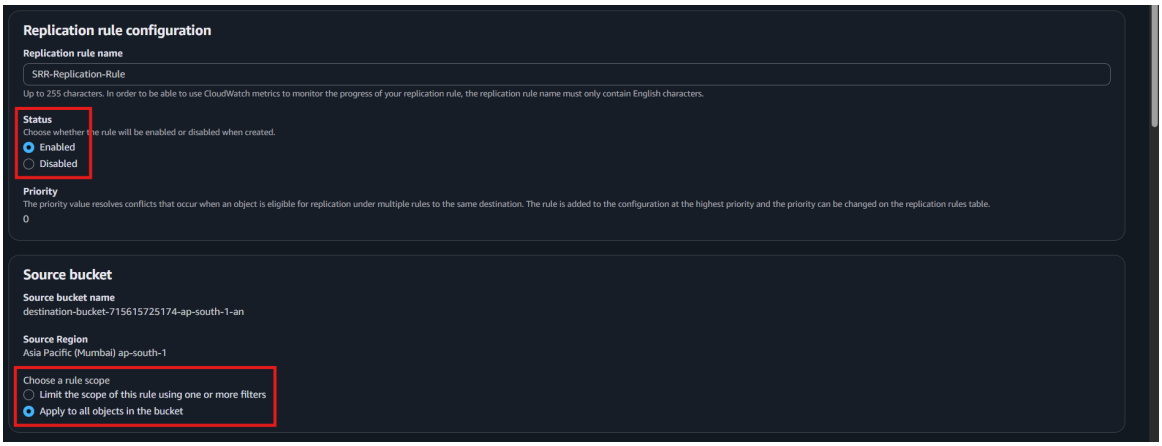


Figure 15: Same-account SRR configuration evidence

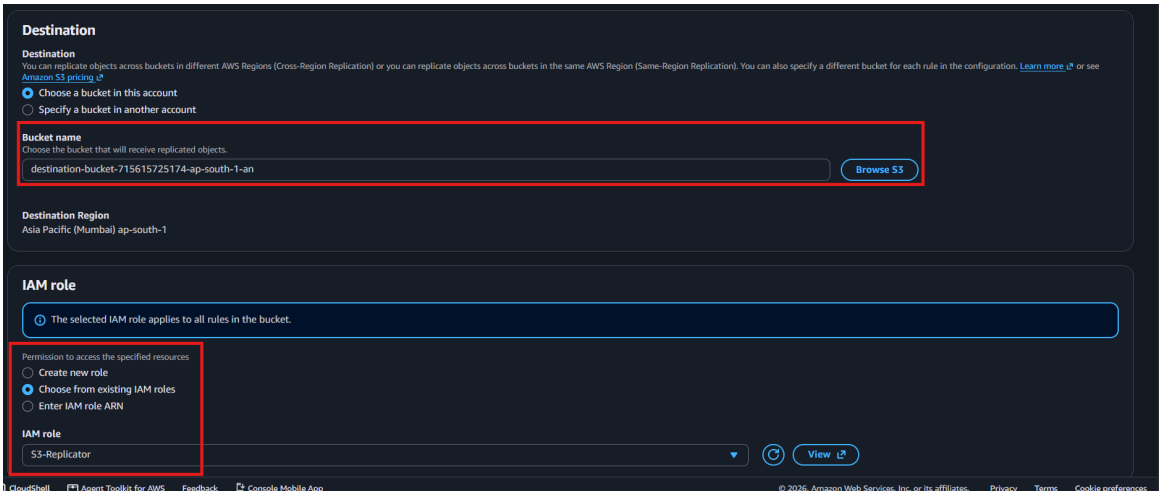


Figure 16: Same-account SRR destination or rule evidence

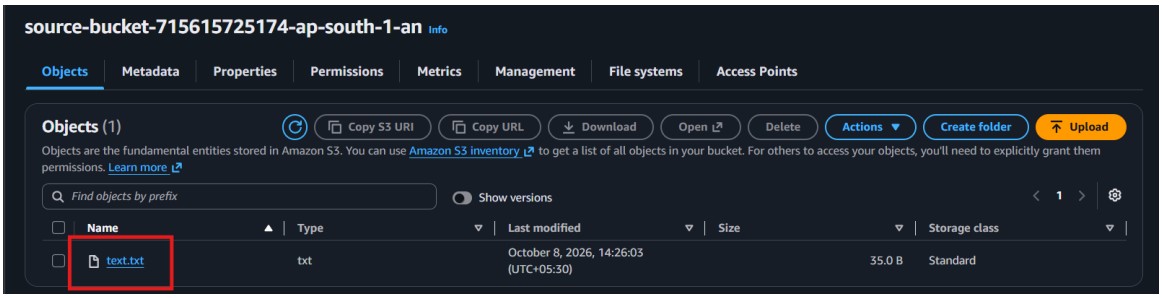


Figure 17: Object or replication status evidence for the SRR scenario

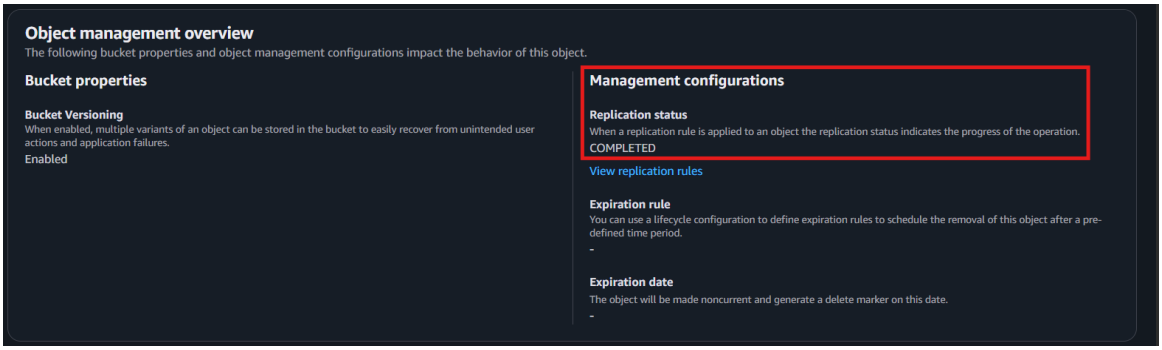


Figure 18: Additional same-account replication evidence

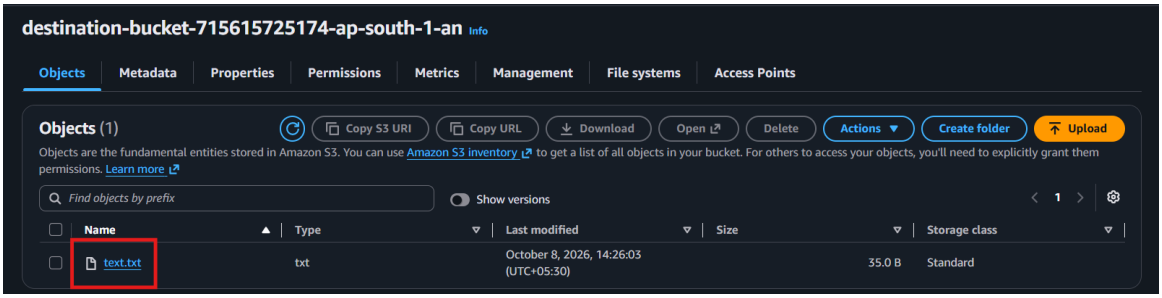


Figure 19: Destination-side verification for the SRR scenario

Verification note: The source and destination must be distinct buckets. If a screenshot appears to show the same bucket selected as both source and destination, it should not be used as proof of a valid source-to-destination replication configuration.

6.2 Scenario 2: Same-Account CRR

This scenario copies eligible objects from Mumbai to Singapore within the same AWS account.

- Source Region: `ap-south-1` (Mumbai)
- Destination Region: `ap-southeast-1` (Singapore)
- Source account: `715615725174`
- Destination account: `715615725174`

- Destination bucket: `s3-singapore-bucket-715615725174-ap-southeast-1-an`

This is cross-Region replication because the buckets are in different AWS Regions, even though they belong to the same account.

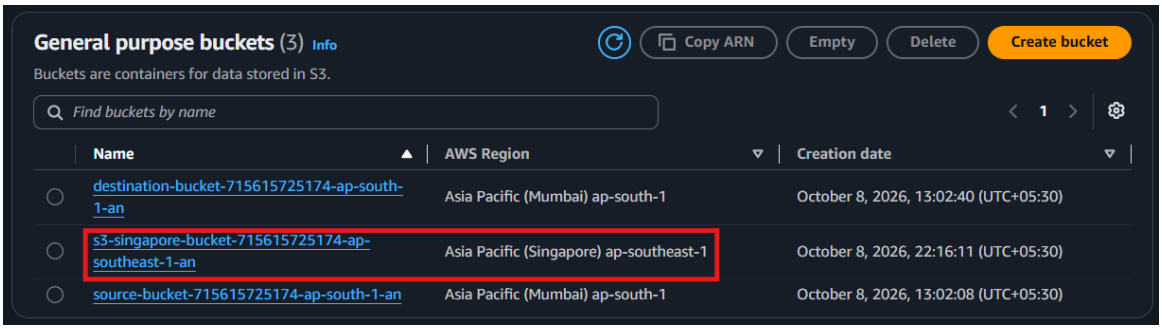


Figure 20: Same-account CRR rule configuration

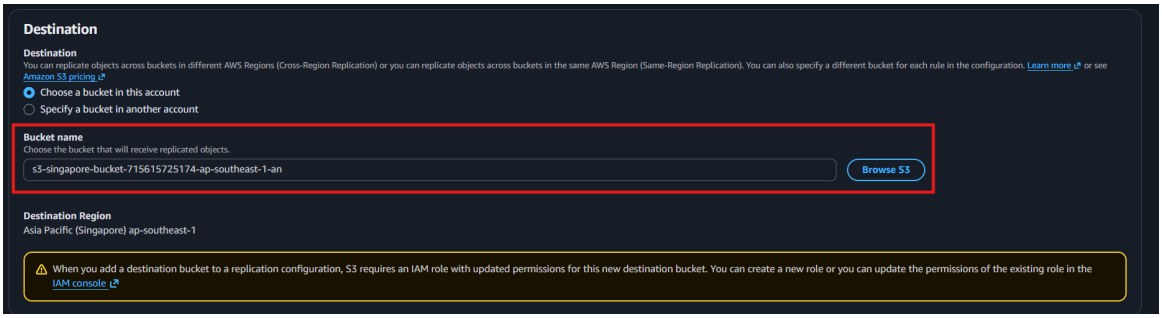


Figure 21: S3-CRR rule status or configuration

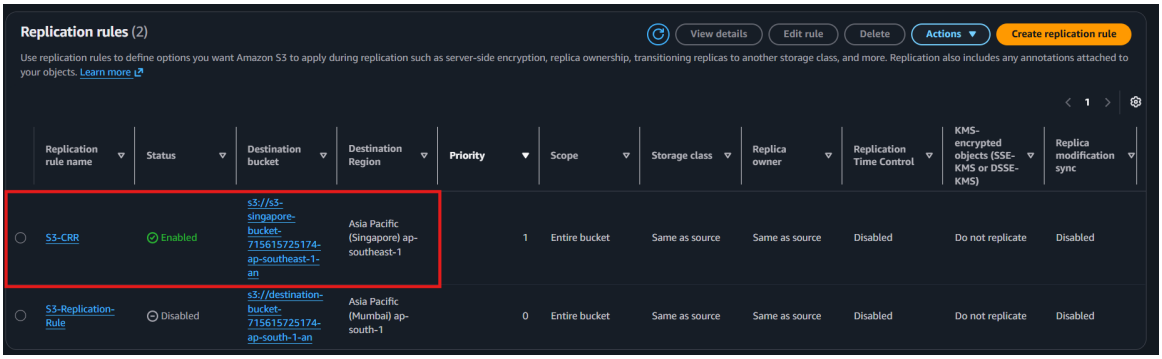


Figure 22: Replication status evidence for the CRR scenario

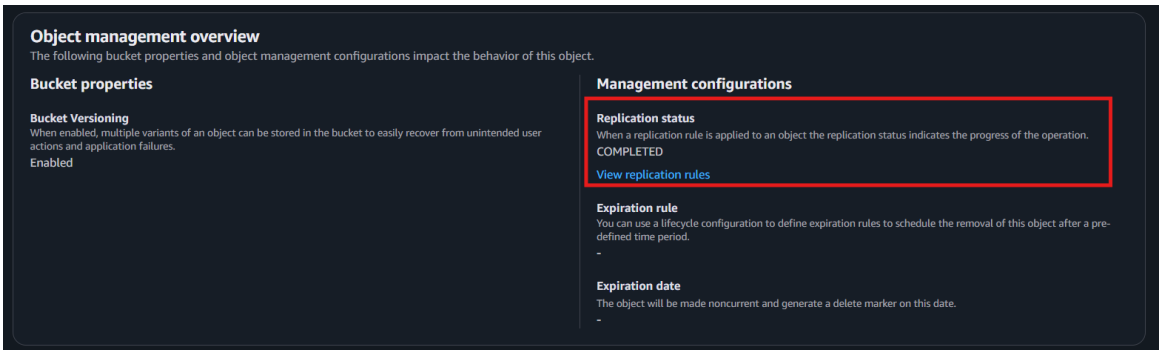


Figure 23: Singapore destination bucket

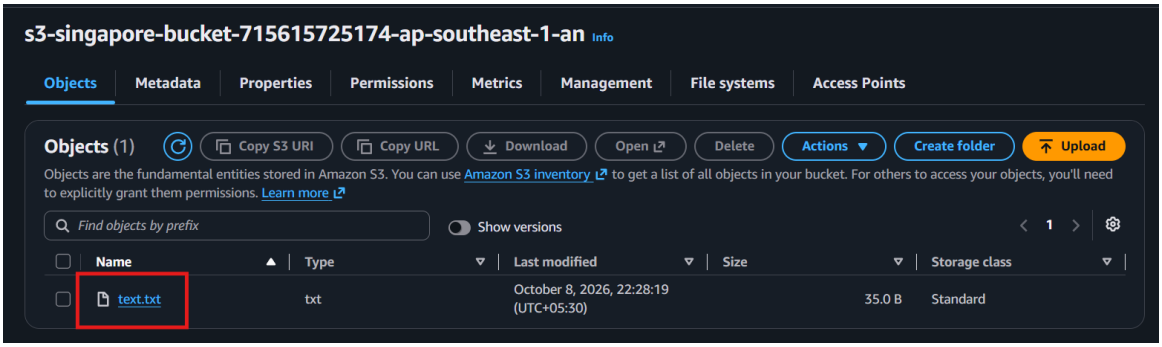


Figure 24: Object evidence in the Singapore destination

6.3 Scenario 3: Cross-Account SRR

In this scenario, the source bucket belongs to account `715615725174`, while the destination bucket belongs to account `342572234319`. Both buckets are in the Mumbai Region.

- Source Region: `ap-south-1`
- Destination Region: `ap-south-1`
- Destination bucket: `destination-bucket-342572234319-ap-south-1-an`
- Destination account: `342572234319`

Because the destination is in another AWS account, its bucket policy must allow the source account's replication role to perform the required replication operations.

The destination bucket policy is configured on the destination bucket in the destination account. Its principal identifies the source account's `S3-Replicator` role.

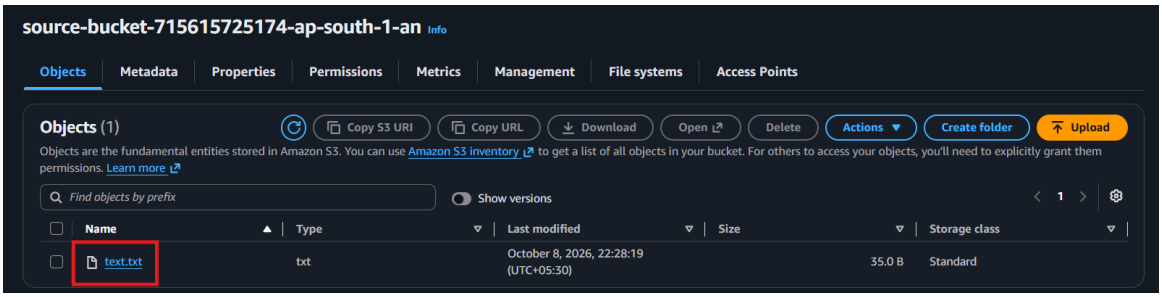


Figure 25: Cross-account SRR source object or source bucket

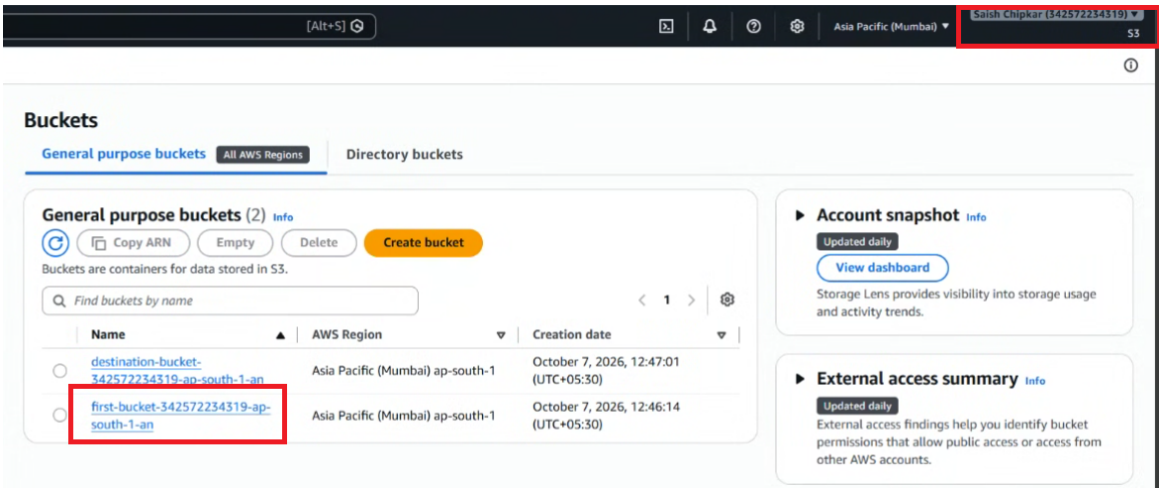


Figure 26: Cross-account SRR destination bucket

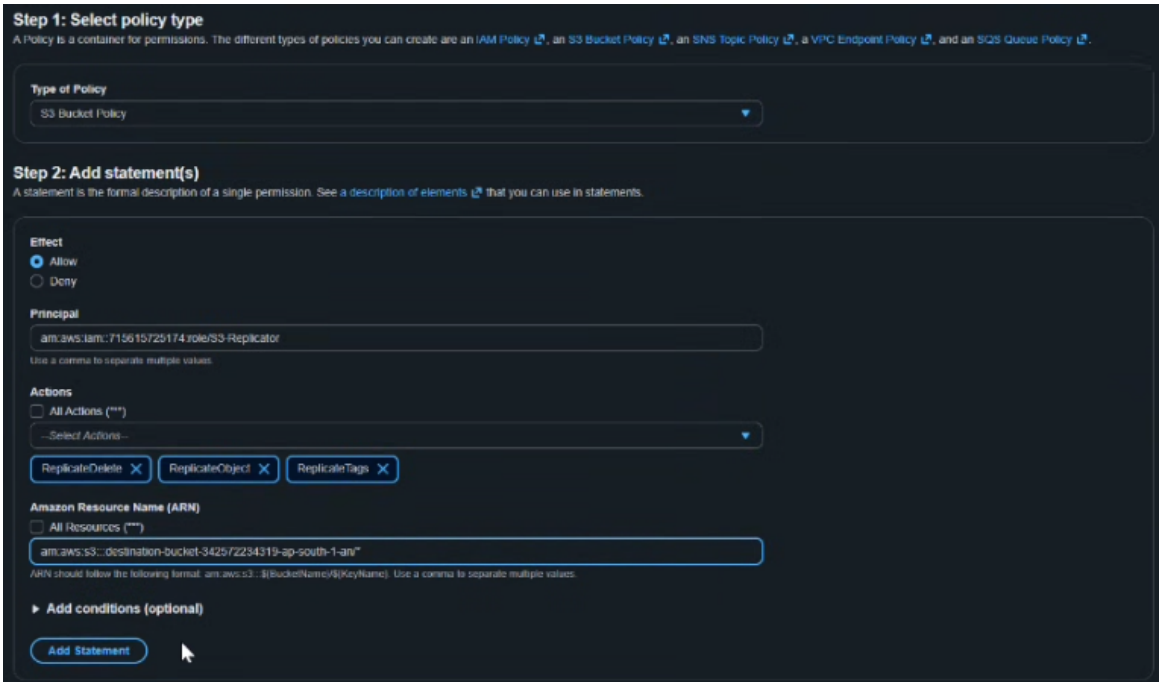


Figure 27: Destination bucket policy editor

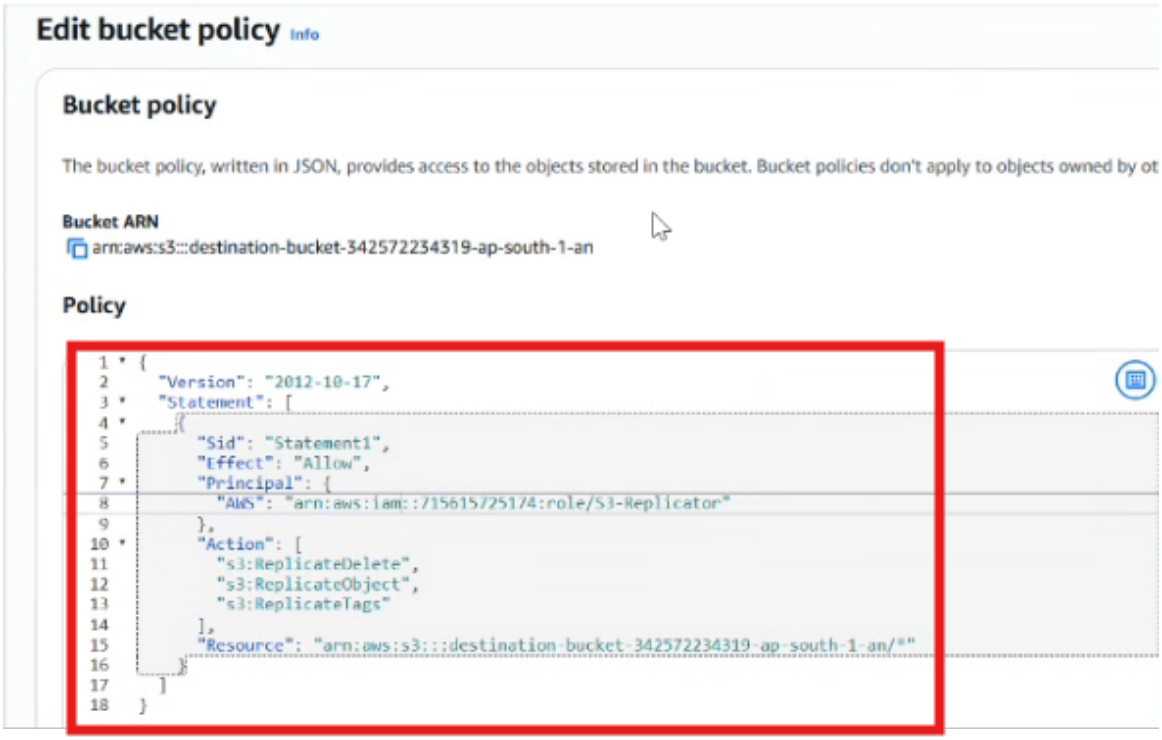


Figure 28: Cross-account destination bucket policy configuration

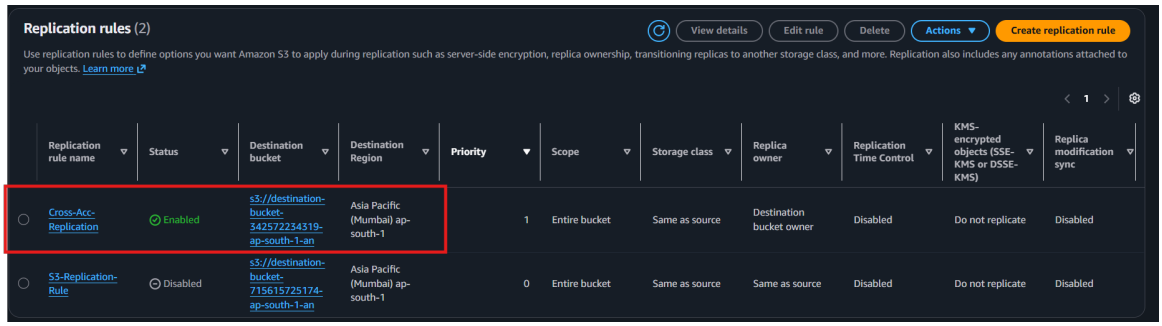


Figure 29: Cross-account SRR replication rule

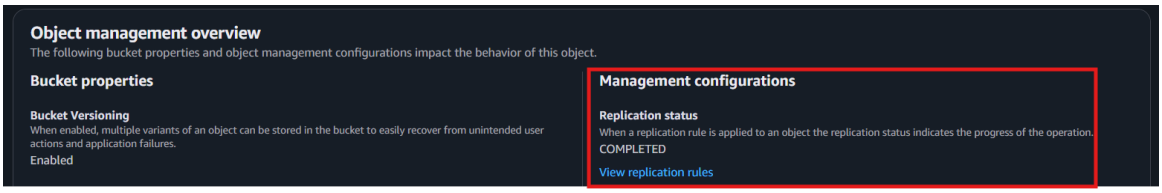


Figure 30: Replication status evidence

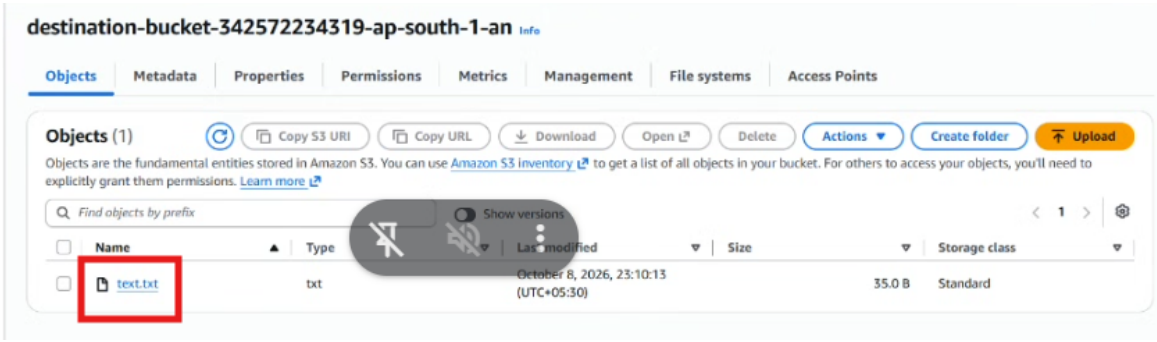


Figure 31: Object verification in the cross-account destination

6.4 Scenario 4: Cross-Account CRR

This scenario combines cross-account and cross-Region replication.

- Source account: `715615725174`
- Source Region: `ap-south-1` (Mumbai)
- Destination account: `342572234319`
- Destination Region: `us-east-2` (Ohio)
- Destination bucket: `amz-cross-region-342572234319-us-east-2-an`

The destination bucket belongs to another account and is located in a different Region. Therefore, the source role's permissions and the destination bucket policy must both authorize the intended replication operations.

The configured rule excluded SSE-KMS and DSSE-KMS encrypted objects. The test object used in this experiment was reported as SSE-S3 encrypted, so that KMS exclusion does not itself prevent replication of that test object.

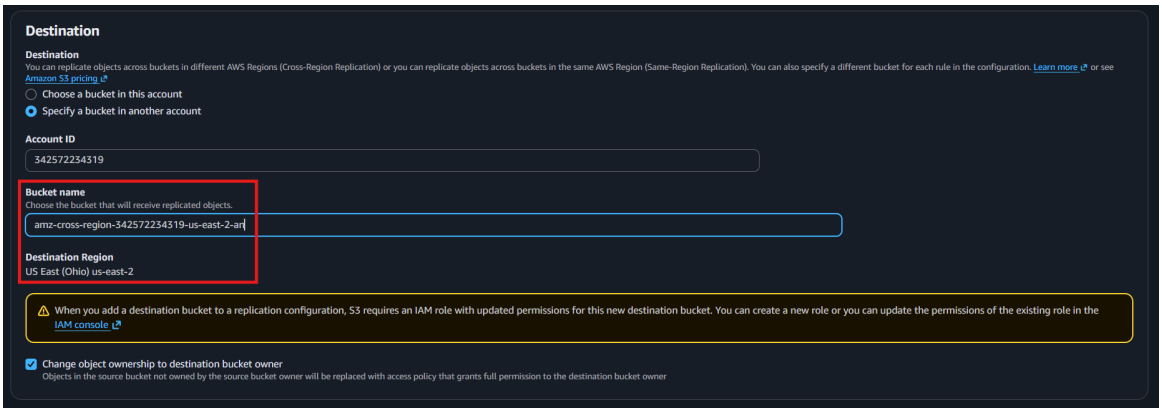


Figure 32: Ohio destination and cross-account CRR setup

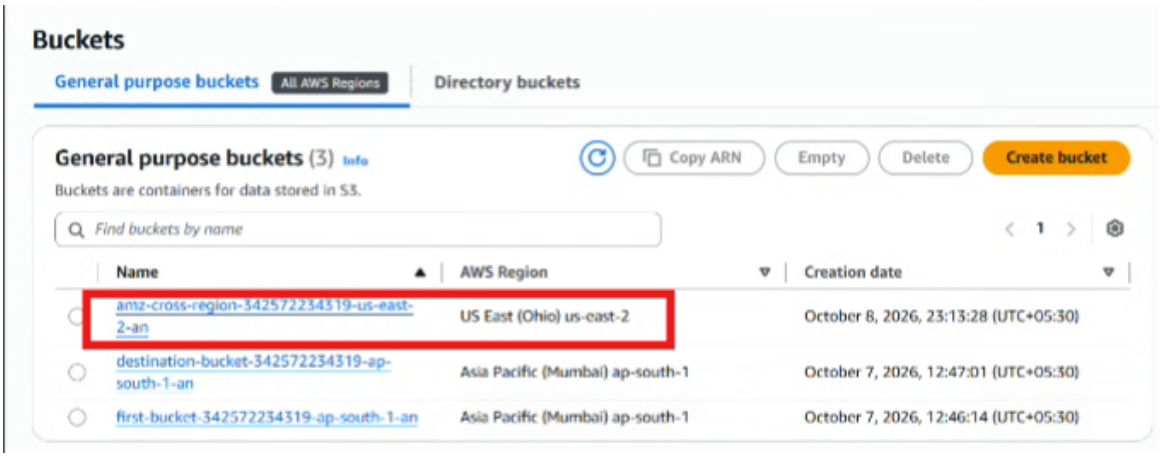


Figure 33: Source object for the cross-account CRR test

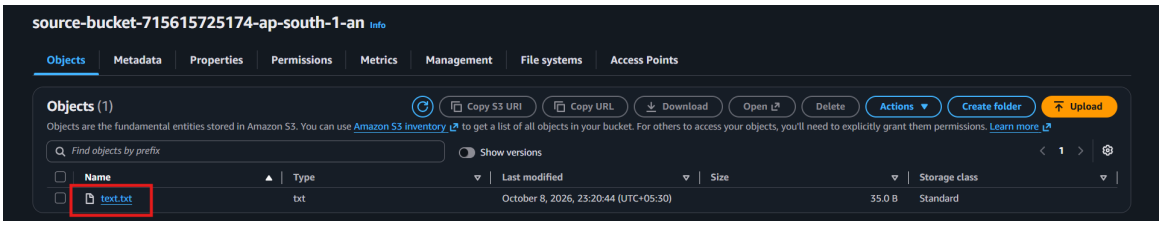


Figure 34: Cross-account CRR configuration or replication status

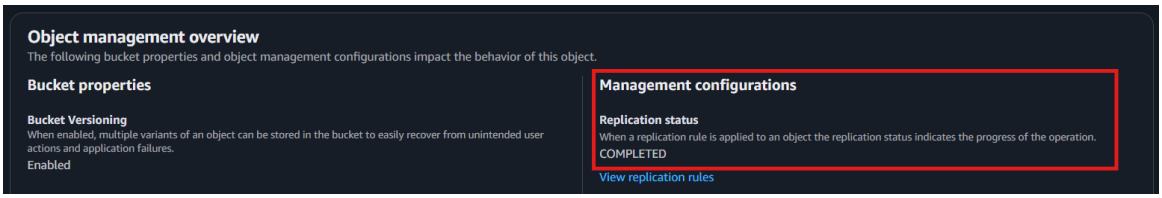


Figure 35: Replication completion evidence

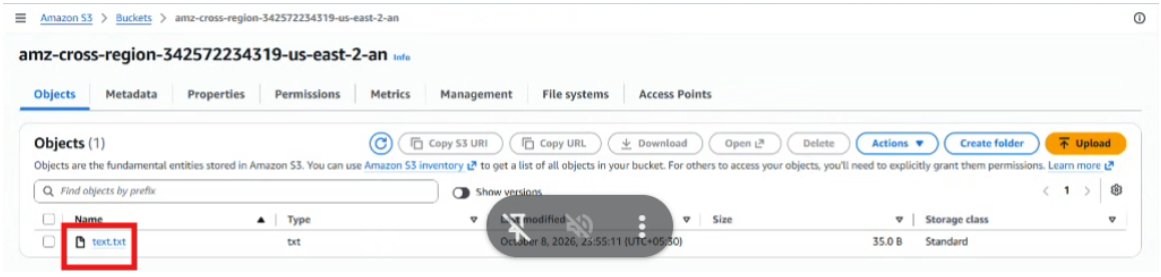


Figure 36: Object verification in the Ohio destination bucket

7 Cross-Account Permissions

For cross-account replication, permissions are configured on both sides.

7.1 Source Account

The source account contains the `S3-Replicator` IAM role. The role's permissions policy grants source-reading permissions and replication permissions for the configured destination object ARN.

7.2 Destination Account

The destination bucket policy must authorize the source role to write replicated objects and perform the other configured replication actions.

Where ownership transfer is configured, the policy may also need `s3:ObjectOwnerOverrideToBucketOwner`

The experiment used the setting that transfers replicated object ownership to the destination bucket owner.

The destination policy should be attached to the destination bucket whose objects are being replicated. A policy for one destination bucket does not automatically grant access to another destination bucket.

7.3 Ownership and Encryption

The configured ownership option was to transfer replicated object ownership to the destination bucket owner.

The source test object was approximately 35 bytes and used SSE-S3 encryption. The Ohio CRR configuration excluded KMS-encrypted objects. The report therefore does not claim that SSE-KMS or DSSE-KMS replication was tested.

8 Verification and Observations

The following checks should be used to evaluate the experiment.

1. **Lifecycle rules:** Confirm the rule names, status, expiration settings, transition settings, and minimum object size.
2. **Versioning:** Confirm that versioning is enabled on each source and destination bucket used for replication.
3. **IAM role:** Confirm that the role trusts Amazon S3 and has the required permissions for the configured source and destination.
4. **Destination policy:** For cross-account replication, confirm that the destination bucket policy authorizes the source replication role.
5. **Replication rule:** Confirm that the correct source bucket and distinct destination bucket are selected.
6. **Object status:** Check replication status in the S3 console where available and check the destination bucket for the test object.

7. **Timing:** Allow for asynchronous replication and lifecycle processing before concluding that an operation failed.

8.1 Important Limitations

- Live replication generally applies to eligible objects created after the replication rule is enabled. Existing objects require a separate approach, such as S3 Batch Replication, if they need to be replicated.
- Lifecycle actions are asynchronous and are not guaranteed to occur immediately after a rule is enabled.
- The 35-byte test object is smaller than the configured 128 KB transition minimum. Its transition to Standard-IA should not be assumed.
- A replication rule's existence is not by itself proof that an object was successfully replicated.
- The supplied IAM role policy references the Ohio destination bucket. Permissions for other destination buckets must be checked against their actual resource ARNs.

9 Conclusion

This experiment covered two important Amazon S3 capabilities: lifecycle management and replication.

Lifecycle rules were configured for noncurrent-version expiration, expired delete-marker removal, incomplete multipart upload cleanup, and transition of eligible objects to S3 Standard-IA.

The replication work covered same-account SRR, same-account CRR, cross-account SRR, and cross-account CRR. The experiment also involved an IAM role trusted by Amazon S3, a role permissions policy, and destination bucket permissions for cross-account replication.

The key lesson is that successful configuration depends on more than creating a rule. Versioning, resource-specific IAM permissions, destination bucket policies, object eligibility, encryption settings, and asynchronous processing must all be considered when verifying the result.

A Appendix A: IAM User Policy

The following is the user policy supplied for this experiment. It includes permissions used by this and other S3 exercises. The presence of a permission in this final policy does not necessarily mean it was newly added for this experiment.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "VisualEditor0",
      "Effect": "Allow",
      "Action": "iam:PassRole",
```

```
      "Resource": "arn:aws:iam::715615725174:role/S3-Replicator"
    },
    {
      "Sid": "VisualEditor1",
      "Effect": "Allow",
      "Action": [
        "s3:GetLifecycleConfiguration",
        "s3:ListBucketVersions",
        "s3:CreateBucket",
        "s3:ListBucket",
        "s3:GetBucketPolicy",
        "s3:PutEncryptionConfiguration",
        "s3:GetObjectAcl",
        "s3:PutLifecycleConfiguration",
        "access-analyzer:ValidatePolicy",
        "s3:DeleteObject",
        "s3:PutBucketVersioning",
        "s3:PutObjectAcl",
        "s3:GetBucketPublicAccessBlock",
        "s3:PutBucketPublicAccessBlock",
        "s3:GetBucketWebsite",
        "s3:PutBucketOwnershipControls",
        "s3:PutReplicationConfiguration",
        "iam:ListRoles",
        "s3:GetBucketVersioning",
        "s3:GetBucketAcl",
        "s3:GetReplicationConfiguration",
        "s3:PutObject",
        "s3:GetObject",
        "s3:PutBucketWebsite",
        "s3:ListAllMyBuckets",
        "s3:PutBucketPolicy",
        "s3:GetBucketLocation",
        "s3:GetObjectVersion"
      ],
      "Resource": "*"
    }
  ]
}
```

Security note: The general S3 statement uses `"Resource": "*"` as supplied. This is broad access and should be reviewed before using the policy outside a learning environment. Where practical, permissions should be scoped to the required bucket and object resources.

B Appendix B: S3-Role-Policy

The following is the supplied permissions policy for the `S3-Replicator` role.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "VisualEditor0",
      "Effect": "Allow",
      "Action": [
        "s3:GetObjectVersionAcl",
        "s3:ListBucket",
```

```
        "s3:GetObjectVersionForReplication",
        "s3:GetBucketVersioning",
        "s3:GetReplicationConfiguration",
        "s3:GetObjectVersion"
      ],
      "Resource": [
        "arn:aws:s3:::source-bucket-715615725174-ap-south-1-an",
        "arn:aws:s3:::source-bucket-715615725174-ap-south-1-an/*"
      ]
    },
    {
      "Sid": "VisualEditor1",
      "Effect": "Allow",
      "Action": [
        "s3:ReplicateObject",
        "s3:ObjectOwnerOverrideToBucketOwner",
        "s3:ReplicateTags",
        "s3:ReplicateDelete"
      ],
      "Resource": "arn:aws:s3:::amz-cross-region-342572234319-us-east-2-an/*"
    }
  ]
}
```

Important: The destination ARN in this supplied policy is the Ohio cross-account CRR bucket. It should not be assumed to cover the Mumbai or Singapore destination buckets.

For other scenarios, review the role policy and ensure that the destination replication permissions reference the intended destination bucket. Resource scope should also be checked against the resource types required by each action.

C Appendix C: Cross-Account Bucket Policy Checklist

The destination bucket policy for a cross-account scenario should be checked for the following:

- The policy is attached to the correct destination bucket in account `342572234319`.
- The principal identifies the source account's role: `arn:aws:iam::715615725174:role/S3-Replicat`
- The resource identifies the intended destination objects, commonly using the destination bucket ARN followed by `/*`.
- The actions permit the required replication operations, including `s3:ReplicateObject`, `s3:ReplicateDelete`, and `s3:ReplicateTags` where required.
- If ownership override is configured, verify the corresponding permission `s3:ObjectOwnerOverrideTo`
- The source role policy and destination bucket policy must work together; neither should be assumed to replace the other.

The exact bucket policy should be checked against the version saved in the AWS console. It is not reproduced as exact JSON here because the full destination bucket policy JSON was not included in the written configuration details available for this report.

D Appendix D: Screenshot Index

The screenshot captions in the main report use the working mapping below. Compare these filenames and captions with the original images before submitting the report, particularly where multiple screenshots show adjacent steps.

File	Expected subject
ss1.png–ss5.png	Lifecycle bucket and Test-Rule configuration.
ss6.png–ss8.png	Standard-IA transition rule and lifecycle rule overview.
ss9.png–ss13.png	IAM role, policy, trust relationship, and replication setup.
ss14.png–ss18.png	Same-account SRR configuration and verification.
ss19.png–ss23.png	Same-account CRR to Singapore.
ss24.png–ss30.png	Cross-account SRR and destination bucket policy.
ss31.png–ss35.png	Cross-account CRR to Ohio.